PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Document Metadata

FIELD	VALUE
Pentester Name (Cybersecurity Professional)	Tanvi Gavatre
Program / Batch	Networkwalks Cybersecurity Internship
Date	11th November 2026
Modules Completed	W2-PM1 (Multiple Kali Tools), W2-PM5 (Zenmap Scanning)
Client / Target	1. Networkwalks (secured written permission already) 2. Local Subnet / Virtual LAN Network
Permission Secured from Client?	Yes
Phases Covered	Phase 1:Reconnaissance & Footprinting Phase 2:Scanning & Network Discovery Phase 3-5:In Progress

1. Liability Disclaimer

This report contains confidential technical information gathered during authorized security testing under the Networkwalks Cybersecurity Internship Program. All activities were executed within the approved scope, covering networkwalks.com and the student's local network. Unauthorized distribution or misuse of this security data is strictly prohibited.

2. Introduction

This report documents the practical execution of Week 2 tasks (W2-PM1 and W2-PM5). The primary objective was to perform domain reconnaissance, collect target infrastructure metadata, identify active hosts on the local subnet, and understand network scanning and topology mapping.

3. Tools Used

- Domain Reconnaissance & Footprinting: WHOIS, WhatWeb, WAFW00F, Nslookup, DNSRecon, cURL.
- Network Discovery & Scanning: Nmap, Zenmap (Nmap GUI).
- Operating Systems: Kali Linux and Windows.

4. Activities Performed

4.1 Footprinting & Reconnaissance (Task W2-PM1)

Target: networkwalks.com

- Executed WHOIS queries to collect domain registration details. Identified GoDaddy.com, LLC as the registrar, with domain creation date 06 November 2019 and expiry date 06 November 2027.
- Used DNSRecon to collect DNS records, identifying SOA and NS servers (ns6135.hostgator.com and ns6136.hostgator.com), MX server mail.networkwalks.com, and A record IP address 192.232.216.135.
- Used WhatWeb to identify website technologies, including Apache, WordPress 7.1.4, Bootstrap 7.1.4 and jQuery 3.7.1.
- Performed DNS resolution using Nslookup, confirming that networkwalks.com resolves to 192.232.216.135.
- Executed WAFW00F to identify Web Application Firewall protection. ModSecurity (SpiderLabs) was detected.
- Used cURL with the -I option to inspect HTTP response headers. The website returned HTTP/2 200 OK, with Apache server information and security-related response headers.
- Confirmed that HTTP requests redirect to HTTPS.

4.2 Network Scanning & Topology Mapping (Task W2-PM5)

Target: Local Subnet

- Identified the local network configuration using Windows IP Configuration.
- Initiated an Nmap Ping Scan (-sn) through the Zenmap GUI.
- Scanned 256 IP addresses and identified 2 active hosts:
- 10.228.143.206
- 10.228.143.225
- The scan was completed in approximately 8.06 seconds using Nmap version 7.99.
- Captured scan results and the network topology using Zenmap's graphical interface.

5. Risk Analysis & Impact

- Information Exposure via Passive Reconnaissance: Publicly available WHOIS and DNS records reveal domain and infrastructure information that could assist external reconnaissance.
- Technology Disclosure: Identifying website technologies may help attackers research potential weaknesses in the technology stack.
- Network Visibility: Discovering active hosts provides information about devices connected to the local network.
- No open ports, running services, or vulnerabilities were verified during the Ping Scan

6. Recommendations

1. Keep website software, plugins and server components updated.
2. Review DNS records and domain registration information regularly.
3. Maintain updated Web Application Firewall rules and configurations.
4. Monitor active devices and network traffic regularly.
5. Perform authorized port scanning and vulnerability assessments when required.

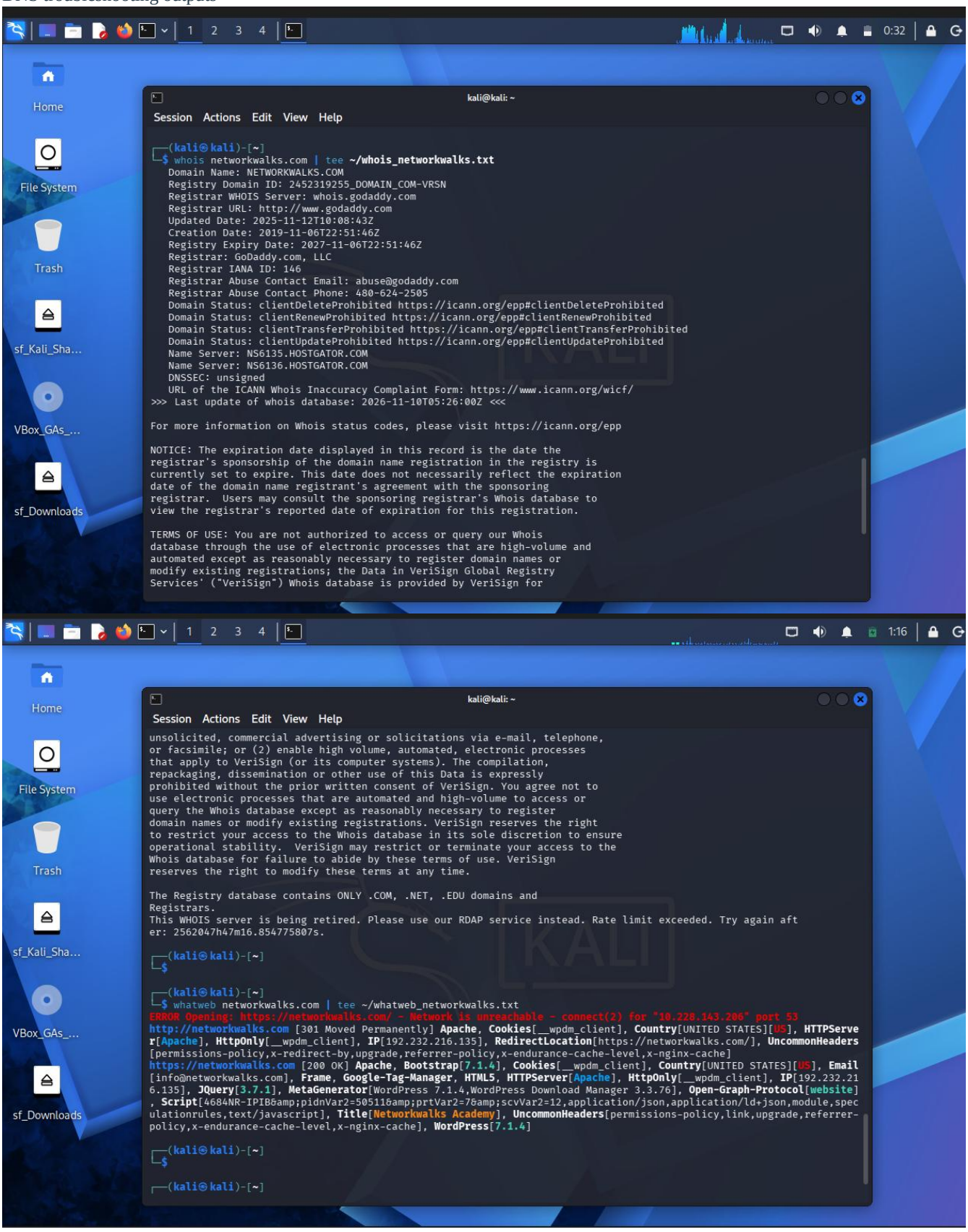
7. Conclusion

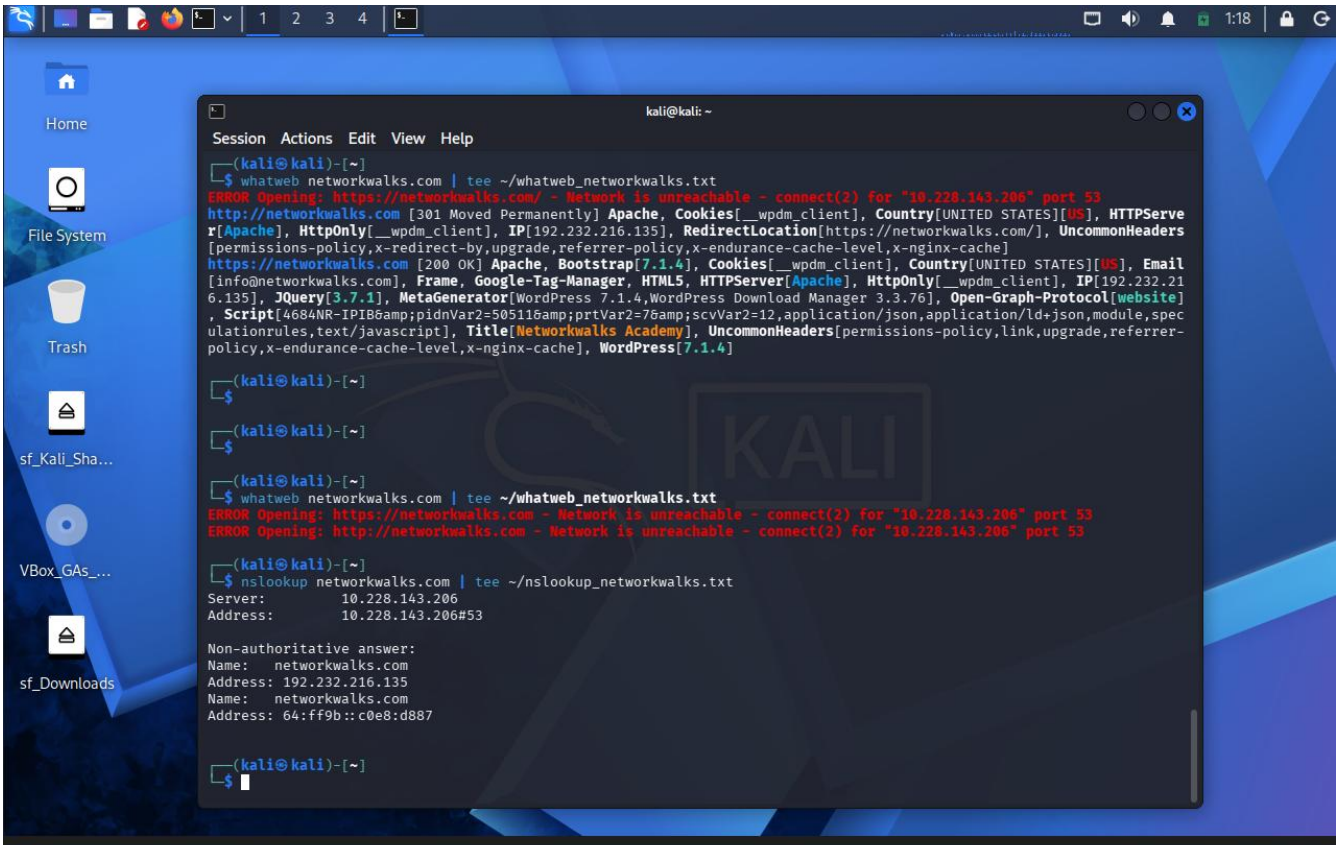
The Week 2 tasks (W2-PM1 and W2-PM5) provided practical experience in domain footprinting, DNS enumeration, website technology detection, network discovery and topology visualization. The collected terminal outputs and screenshots were saved as technical evidence for documentation and internship submission.

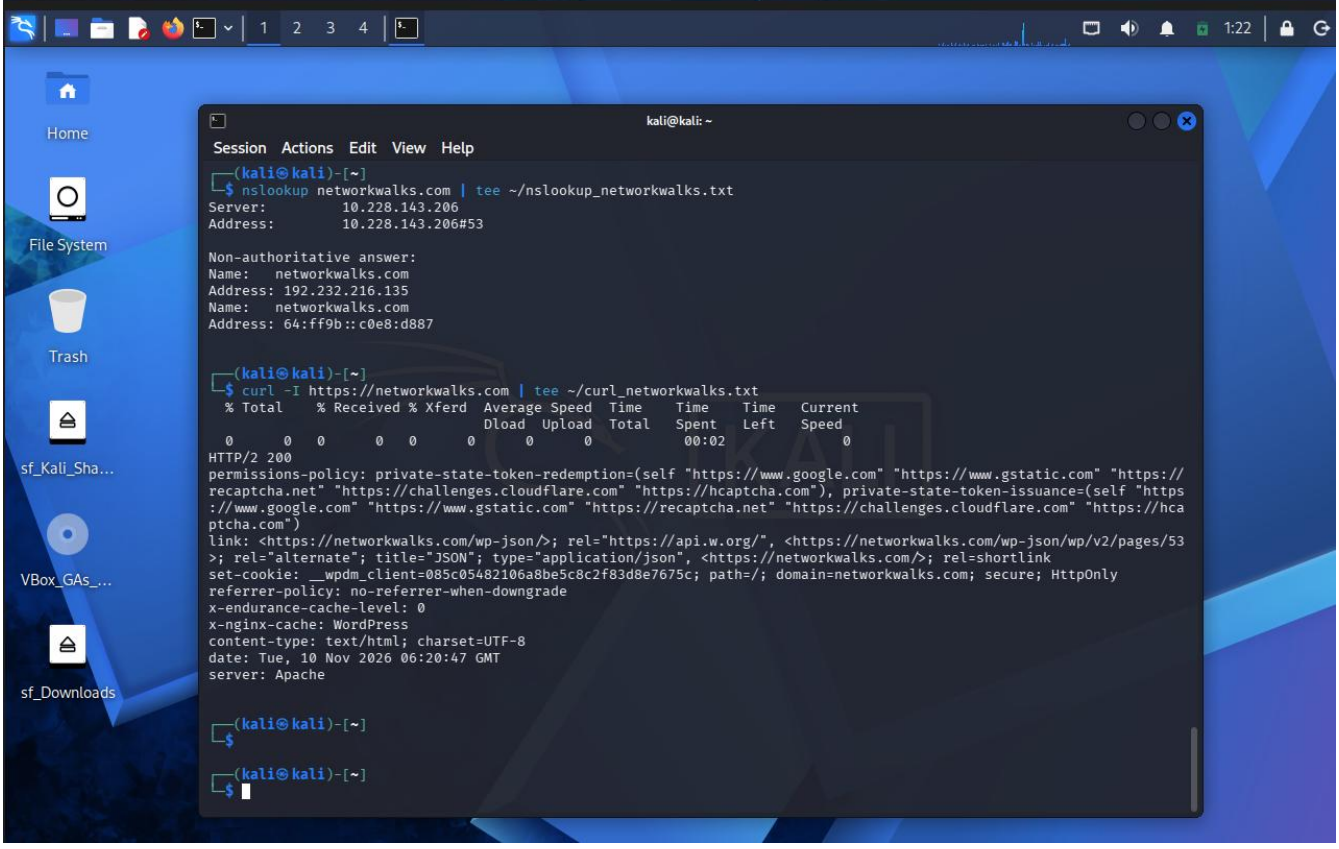
8. Evidences Collected

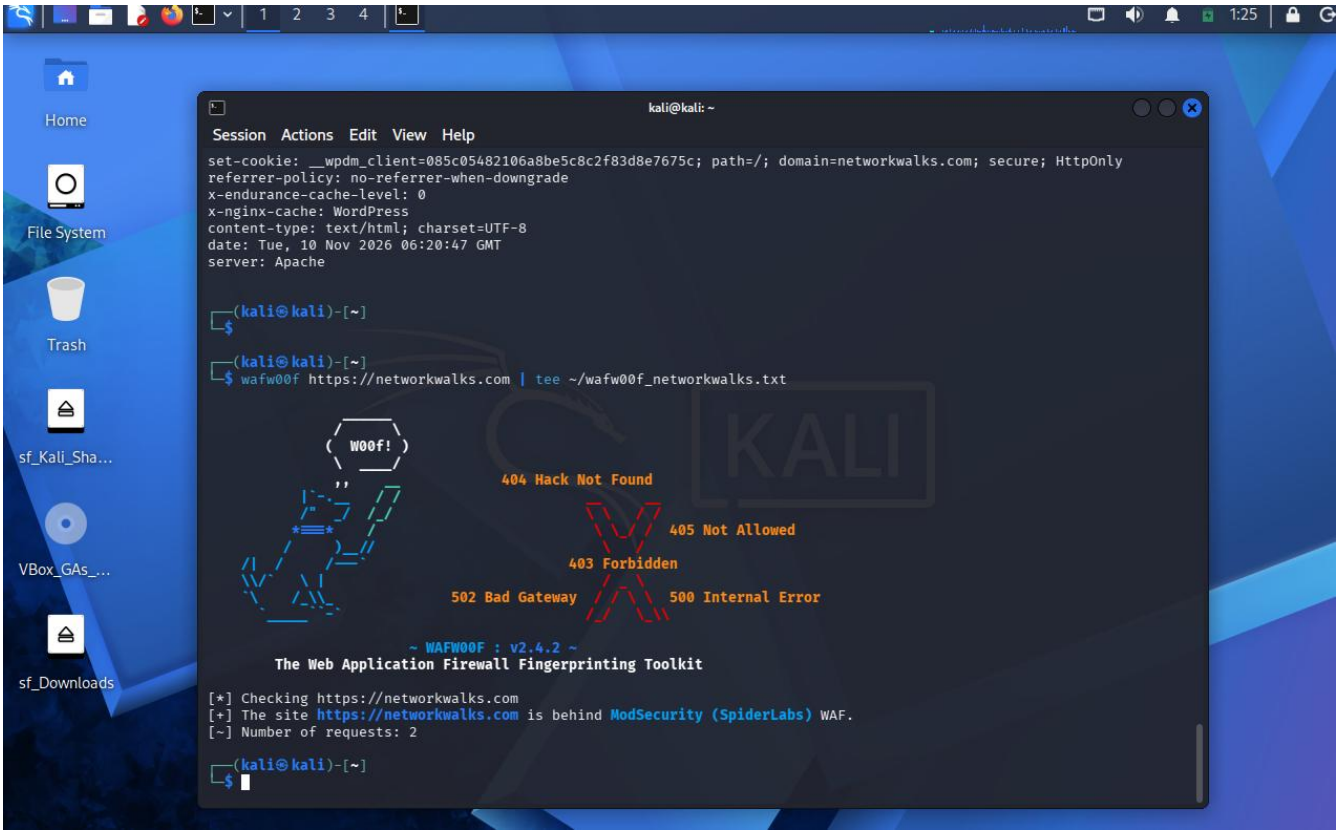
- WHOIS terminal output.
- WhatWeb technology detection output.
- Nslookup DNS resolution results.
- DNSRecon enumeration output.
- cURL HTTP response header output.
- WAFW00F detection output.
- Zenmap Nmap Ping Scan screenshot.
- Zenmap network topology screenshot.

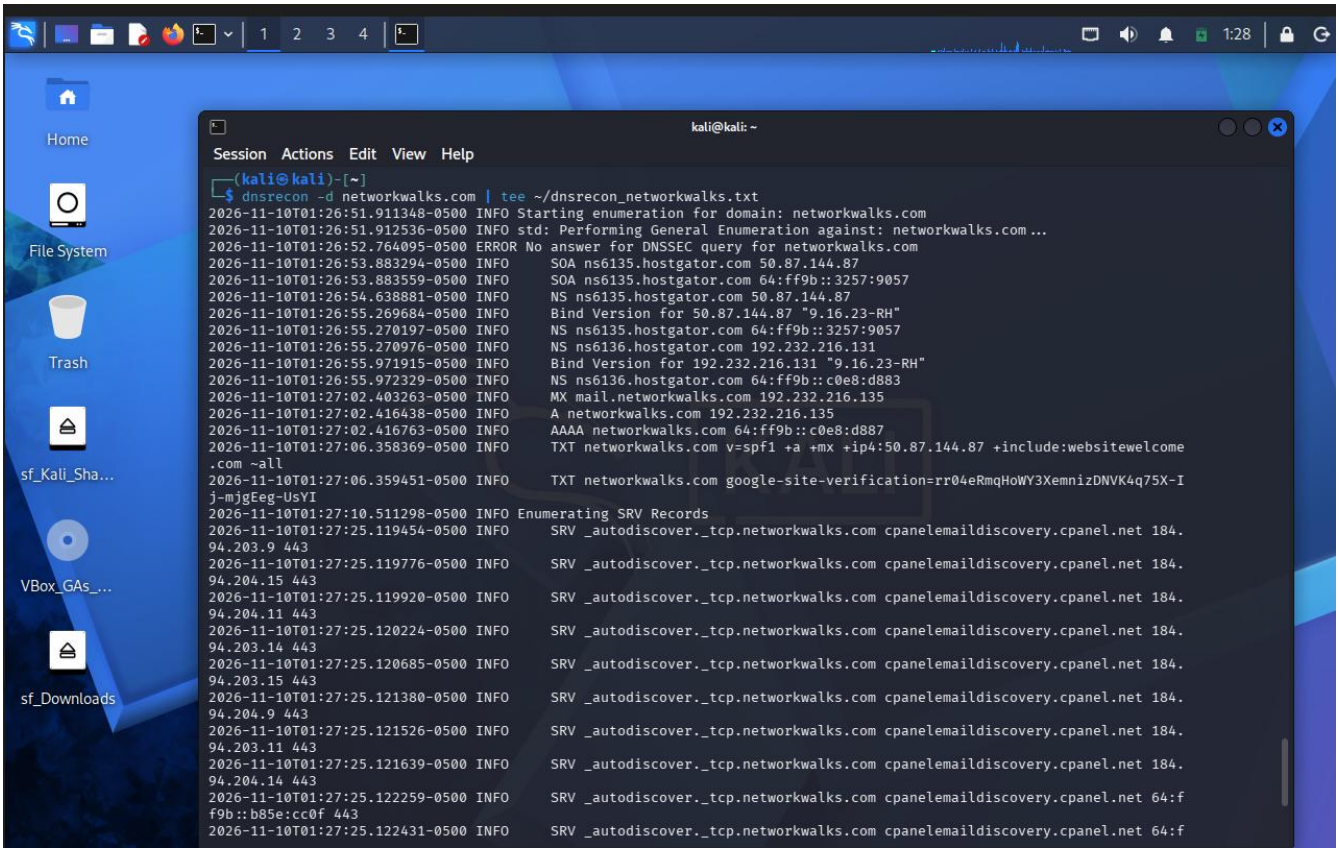
- DNS troubleshooting outputs

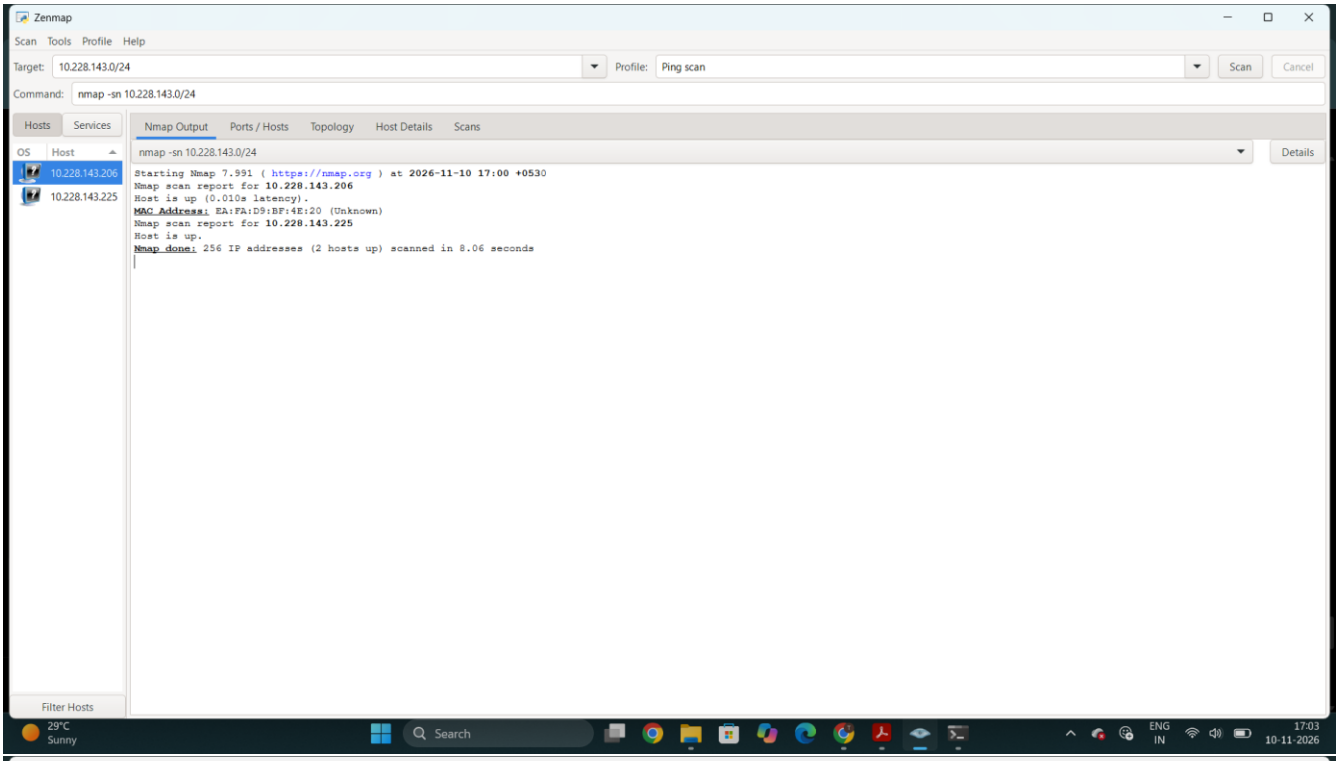
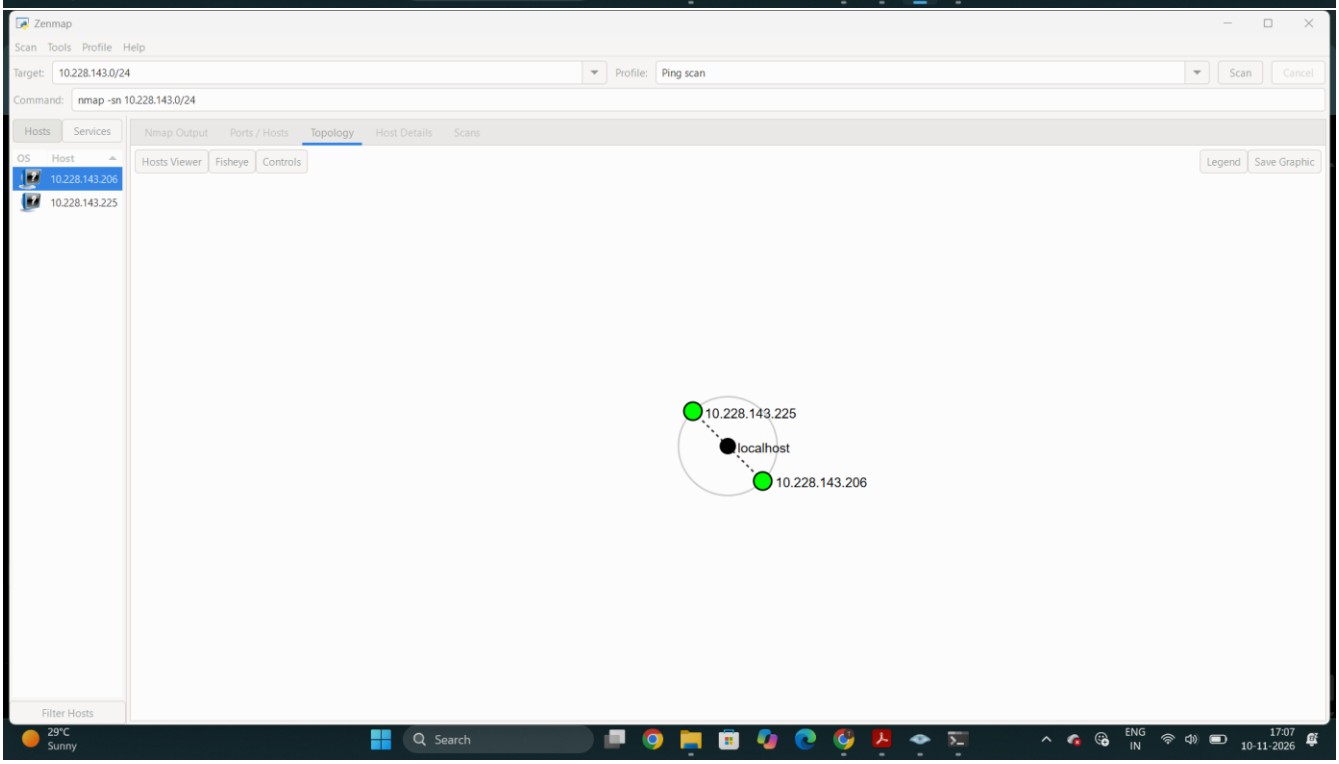



1 2 3 4
1:18
Home
File System
Trash
sf_Kali_Sha...
VBox_GAs_...
sf_Downloads
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ whatweb networkwalks.com | tee ~/whatweb_networkwalks.txt
ERROR Opening: https://networkwalks.com/ - Network is unreachable - connect(2) for "10.228.143.206" port 53
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=505118&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
┌──(kali㉿kali)-[~]
└─$
┌──(kali㉿kali)-[~]
└─$
┌──(kali㉿kali)-[~]
└─$ whatweb networkwalks.com | tee ~/whatweb_networkwalks.txt
ERROR Opening: https://networkwalks.com - Network is unreachable - connect(2) for "10.228.143.206" port 53
ERROR Opening: http://networkwalks.com - Network is unreachable - connect(2) for "10.228.143.206" port 53
┌──(kali㉿kali)-[~]
└─$ nslookup networkwalks.com | tee ~/nslookup_networkwalks.txt
Server: 10.228.143.206
Address: 10.228.143.206#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
Name: networkwalks.com
Address: 64:ff9b::c0e8:d887
┌──(kali㉿kali)-[~]
└─$


1 2 3 4
1:22
Home
File System
Trash
sf_Kali_Sha...
VBox_GAs_...
sf_Downloads
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ nslookup networkwalks.com | tee ~/nslookup_networkwalks.txt
Server: 10.228.143.206
Address: 10.228.143.206#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
Name: networkwalks.com
Address: 64:ff9b::c0e8:d887
┌──(kali㉿kali)-[~]
└─$ curl -I https://networkwalks.com | tee ~/curl_networkwalks.txt
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
0 0 0 0 0 0 0 0 00:02 0
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=085c05482106a8be5c8c2f83d8e7675c; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Tue, 10 Nov 2026 06:20:47 GMT
server: Apache
┌──(kali㉿kali)-[~]
└─$
┌──(kali㉿kali)-[~]
└─$


1 2 3 4
1:25
Home
File System
Trash
sf_Kali_Sha...
VBox_GAs_...
sf_Downloads
kali@kali: ~
Session Actions Edit View Help
set-cookie: __wpdm_client=085c05482106a8be5c8c2f83d8e7675c; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Tue, 10 Nov 2026 06:20:47 GMT
server: Apache
(kali㉿kali)-[~]
$
(kali㉿kali)-[~]
$ wafw00f https://networkwalks.com | tee ~/wafw00f_networkwalks.txt
(W00f!)
404 Hack Not Found
405 Not Allowed
403 Forbidden
502 Bad Gateway
500 Internal Error
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
(kali㉿kali)-[~]
$


1 2 3 4
1:28
Home
File System
Trash
sf_Kali_Sha...
VBox_GAs_...
sf_Downloads
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ dnsrecon -d networkwalks.com | tee ~/dnsrecon_networkwalks.txt
2026-11-10T01:26:51.911348-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-10T01:26:51.912536-0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-10T01:26:52.764095-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-10T01:26:53.883294-0500 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-11-10T01:26:53.883559-0500 INFO SOA ns6135.hostgator.com 64:ff9b::3257:9057
2026-11-10T01:26:54.638881-0500 INFO NS ns6135.hostgator.com 50.87.144.87
2026-11-10T01:26:55.269684-0500 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-10T01:26:55.270197-0500 INFO NS ns6135.hostgator.com 64:ff9b::3257:9057
2026-11-10T01:26:55.270976-0500 INFO NS ns6136.hostgator.com 192.232.216.131
2026-11-10T01:26:55.971915-0500 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-10T01:26:55.972329-0500 INFO NS ns6136.hostgator.com 64:ff9b::c0e8:d883
2026-11-10T01:27:02.403263-0500 INFO MX mail.networkwalks.com 192.232.216.135
2026-11-10T01:27:02.416438-0500 INFO A networkwalks.com 192.232.216.135
2026-11-10T01:27:02.416763-0500 INFO AAAA networkwalks.com 64:ff9b::c0e8:d887
2026-11-10T01:27:06.358369-0500 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome
.com ~all
2026-11-10T01:27:06.359451-0500 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-I
j-mjgEeg-UsYI
2026-11-10T01:27:10.511298-0500 INFO Enumerating SRV Records
2026-11-10T01:27:25.119454-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.
94.203.9 443
2026-11-10T01:27:25.119776-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.
94.204.15 443
2026-11-10T01:27:25.119920-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.
94.204.11 443
2026-11-10T01:27:25.120224-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.
94.203.14 443
2026-11-10T01:27:25.120685-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.
94.203.15 443
2026-11-10T01:27:25.121380-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.
94.204.9 443
2026-11-10T01:27:25.121526-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.
94.203.11 443
2026-11-10T01:27:25.121639-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.
94.204.14 443
2026-11-10T01:27:25.122259-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 64:f
f9b::b85e:cc0f 443
2026-11-10T01:27:25.122431-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 64:f

- 
Zenmap
Scan Tools Profile Help
Target: 10.228.143.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 10.228.143.0/24
Hosts
Services
OS Host
10.228.143.206
10.228.143.225
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -sn 10.228.143.0/24
Details
Starting Nmap 7.991 (https://nmap.org) at 2026-11-10 17:00 +0530
Nmap scan report for 10.228.143.206
Host is up (0.010s latency).
MAC Address: EA:FA:D9:BF:4E:20 (Unknown)
Nmap scan report for 10.228.143.225
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 8.06 seconds
Filter Hosts
29°C
Sunny
Search
ENG
IN
17:03
10-11-2026
- 
Zenmap
Scan Tools Profile Help
Target: 10.228.143.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 10.228.143.0/24
Hosts
Services
OS Host
10.228.143.206
10.228.143.225
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Hosts Viewer
Fisheye
Controls
Legend
Save Graphic
10.228.143.225
localhost
10.228.143.206
Filter Hosts
29°C
Sunny
Search
ENG
IN
17:07
10-11-2026